
AMBIENT
SYSTEMS
Reimagining the future of smart cities

40%

of global CO2 emissions is attributed to buildings

Climate Impact will largely be felt by building owners and their occupants

$260B+ of portfolio losses
(due to climate change)



Climate Risk can reduce asset value by 20% and annual returns by 40%

Ambitious Net Zero Emission Targets Set by COP UNFCCC (est. 1992) and enforced at national level

‘Conference of the Parties’

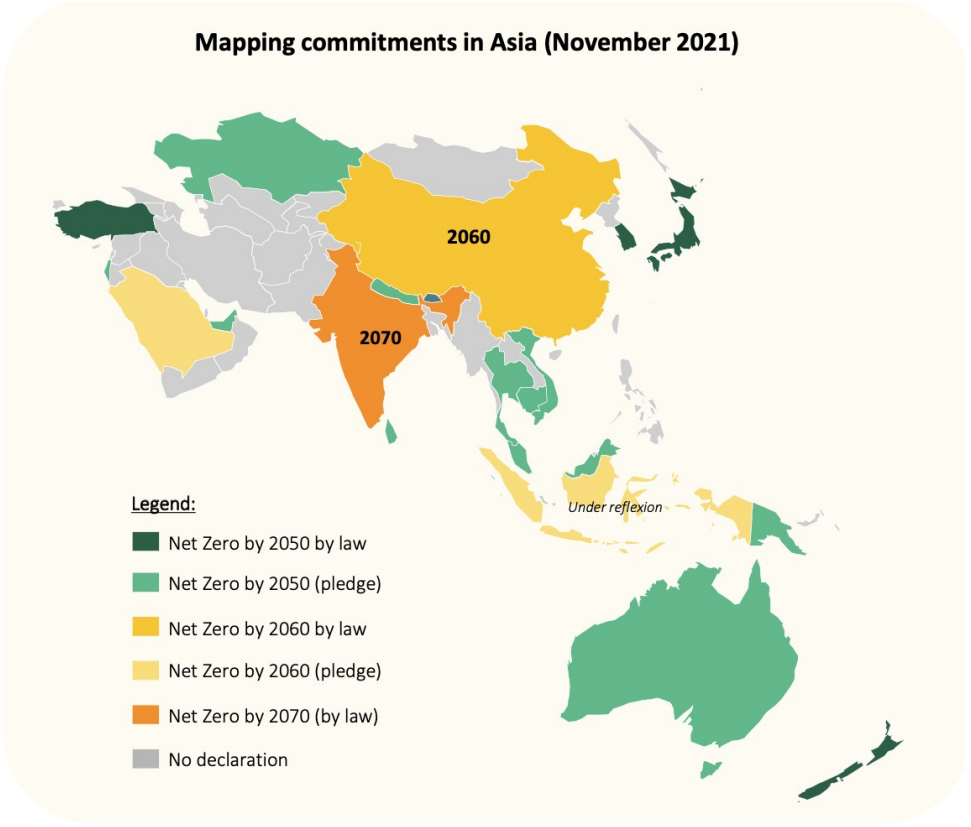
United Nations Framework Convention on Climate Change

May 1992-present

UN CLIMATE CHANGE CONFERENCE UK 2021

IN PARTNERSHIP WITH ITALY





Growing list of Carbon Tracking Methodologies

PCAF Partnership for Carbon Accounting Financials

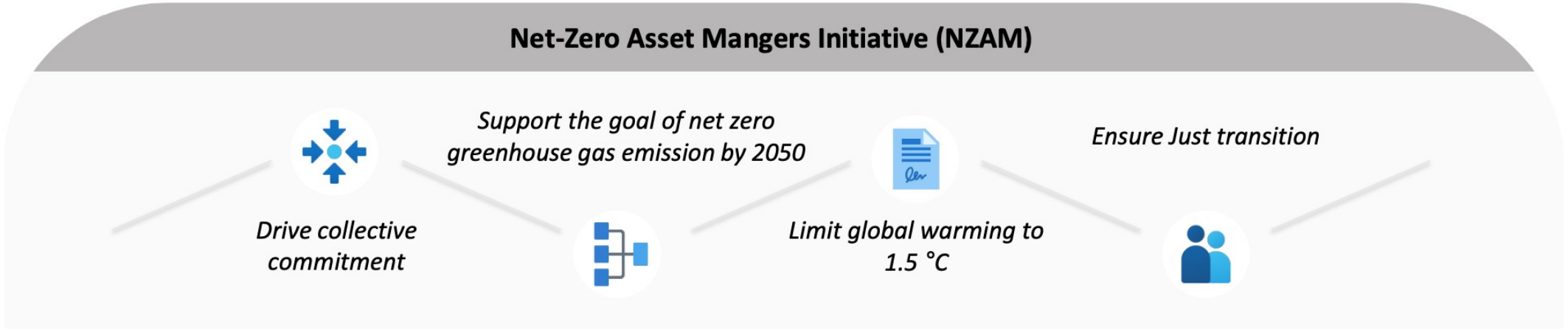
SCIENCE BASED TARGETS
DRIVING AMBITIOUS CORPORATE CLIMATE ACTION

GREENHOUSE GAS PROTOCOL

Global Reporting Initiative™

GRI GSSB Global Sustainability Standards Board

Leading Asset Managers Commit to Net Zero Targets



Commitments

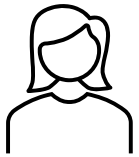
- **Work in partnership with asset owner clients on decarbonisation goals**, consistent with an ambition to reach net zero emissions by 2050 or sooner across all assets under management ('AUM')
- **Set an interim target** for the proportion of assets to be managed **in line with the attainment of net zero emissions by 2050 or sooner**
- **Review the interim target at least every five years**, with a view to ratcheting up the proportion of AUM covered until 100% of assets are included

Personas

Diverse



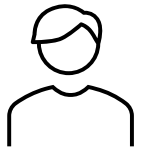
Paying Customer
Primary User





Engineers & Facility Managers

Key Stakeholder
Secondary User
Involved in Buying Process
Contributes Data



End Beneficiary
Contributes Data

PRIMARY USER

Asset Manager & Executives

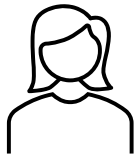


Sample Titles	Managing Director VP, Fund Management Head, Investments Director, Real Estate Finance
Sample Firms	IGIS Asset Management ESR Cayman Blackstone Group Brookfield Asset Management GLP CBRE Investment Management

Key Responsibilities	• Source capital for fund and portfolio • Manage investor relations, including pension funds, fund of funds, and individuals • Compliance & Legal, including financial and ESG • Board and shareholder reporting • Evaluate Transaction • Client Relationship Management (i.e., occupiers)
	• Complex and diverse range of properties and assets • Varying regulatory and operational environments across portfolio countries • Time for Account Management responsibilities with both investors and clients (occupiers)
What they want	• Ways to maximize fund and portfolio valuation and returns • Minimize risks • Focus on relationship building with investors and clients by streamlining technical and time-intensive tasks such as data collection, modelling, and reporting

SECONDARY USER

Property Managers & Brokers



Sample Titles	Property Manager Senior Property Officer
Sample Firms	CapitaLand JLL CBRE

Key Responsibilities

- Manage relationships with occupants
- Renewing leases with key customers
- Preparing reports such as tenants' performance, weekly sales & traffic, monthly management, arrears management, GTO rent billing
- Involved in budgeting and budget tracking
- Multiple communication channels with tenants

What they want

- Simpler communication channels with tenants, asset managers, and contractors

Engineers & Facility Management Staff



Sample Titles	Facility Manager Facilities Operations Facilities Engineer
Sample Firms	CPG Facilities Management CBRE Workplace Solutions Savills Singapore EM Services JLL Engie Facilities Management

Key Responsibilities

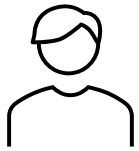
- Oversee a team of technicians and maintenance staff
- Oversee contractors on new projects and retrofits
- Manage relationships with contractors and vendors
- Manage relationships with occupants
- Involved in budgeting and budget tracking
- Involvement in the development and operation of leasing related IT systems such as lease management and business intelligence systems

- Finding relevant personnel in a timely manner
- Expensive labor and contractor costs
- Complex technical repair and maintenance
- Multiple communication channels with tenants

What they want

- More predictable workload
- Simpler communication channels with tenants, asset managers, and contractors
- Ways to ensure quality of maintenance & repair activities
- Minimize risk associated with safety and maintenance

Client Occupants



Sample Titles

General Manager
Operations Manager
Lease Manager
Lease Administration Manager

Sample Firms

Marks & Spencer
H&M
Google

Key Responsibilities

- Manages space requirements for the company
- Identifies suitable locations and preferred amentias
- Negotiates leasing terms with the property manager
- Signs commercial lease for the company
- Sometimes responsible for P&L of the company or business unit

- May have to manage many facilities across different regions
- Lease management one of many other high priority responsibilities

What they want

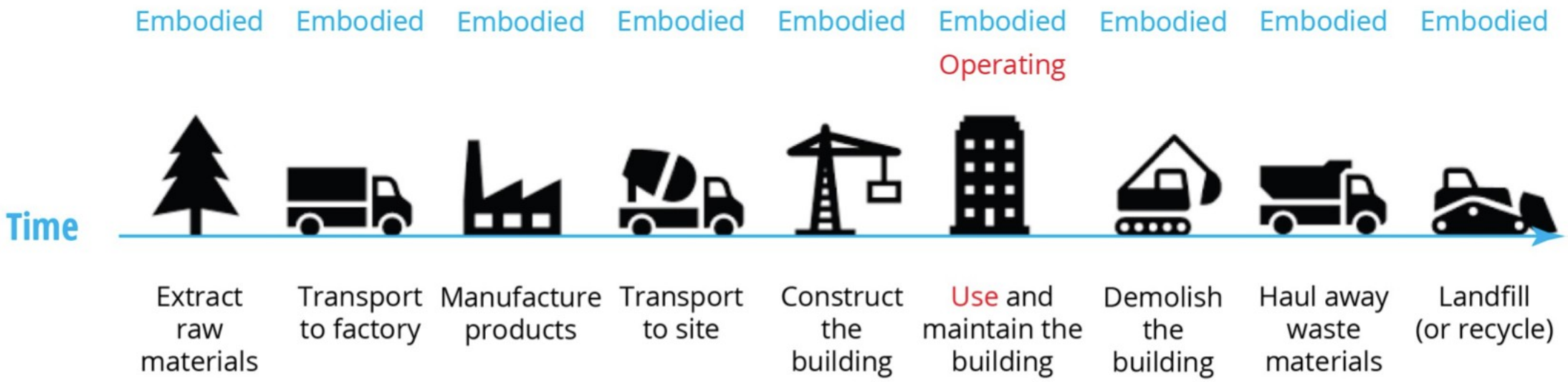
- Attractive lease terms, cost, amenities, duration
- Simple property evaluation process

Product Offering

Real Estate's Most Intelligent Decarbonization Platform

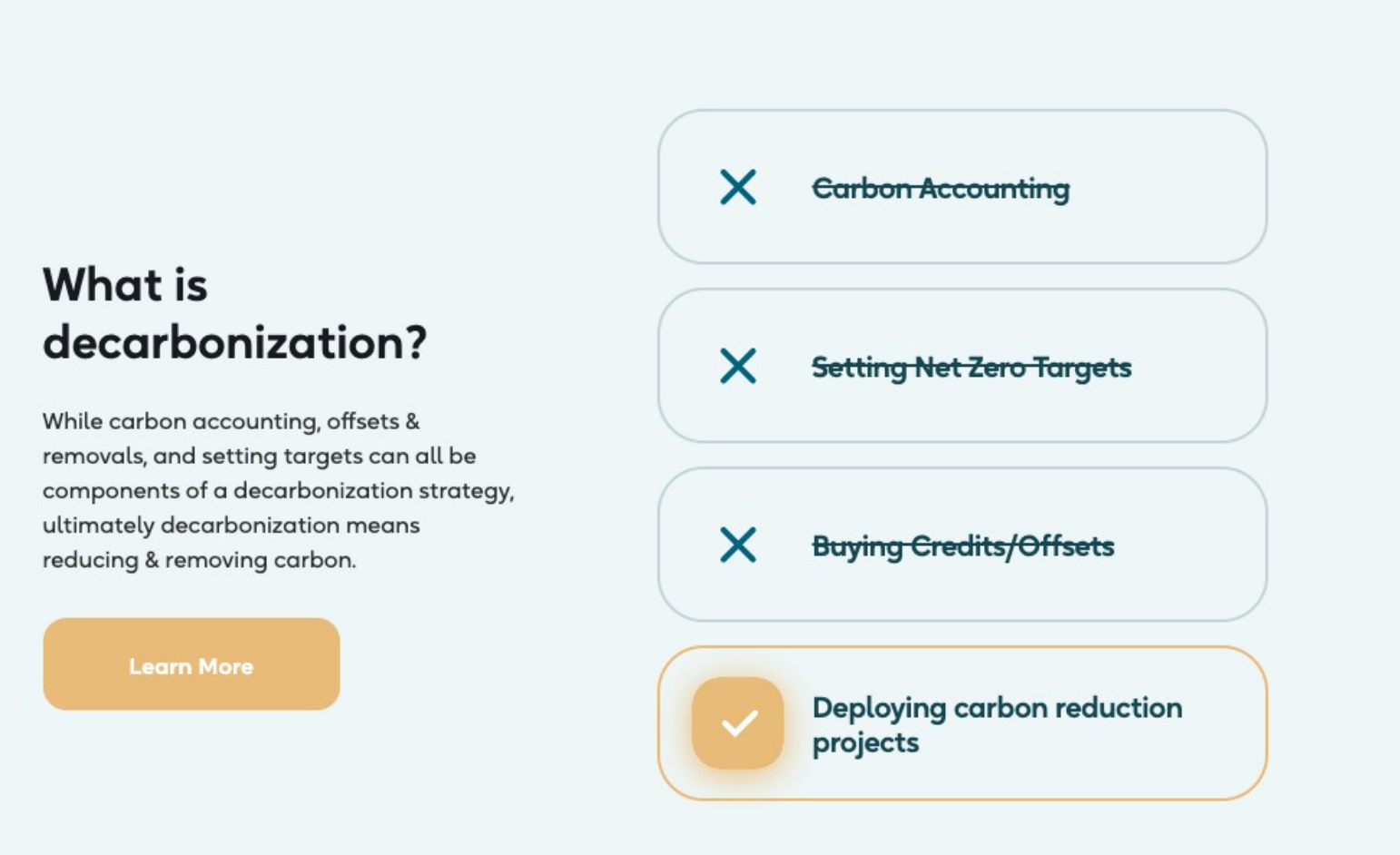
Streamlined management of property operations with Ambient's state of the art platform and proprietary data science methodology

Decarbonization plans across the building lifecycle



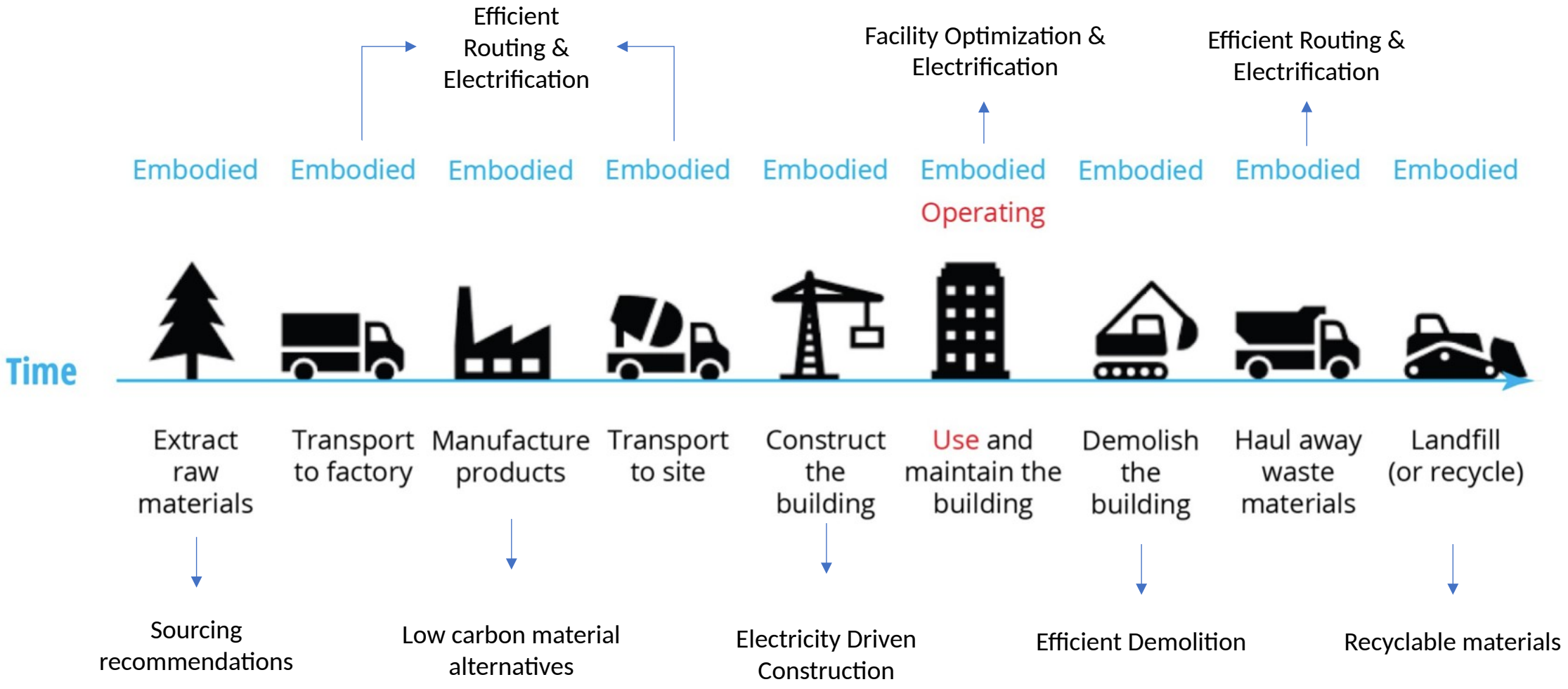
Source: ASHRAE

Domain expertise in decarbonization techniques specific for real estate

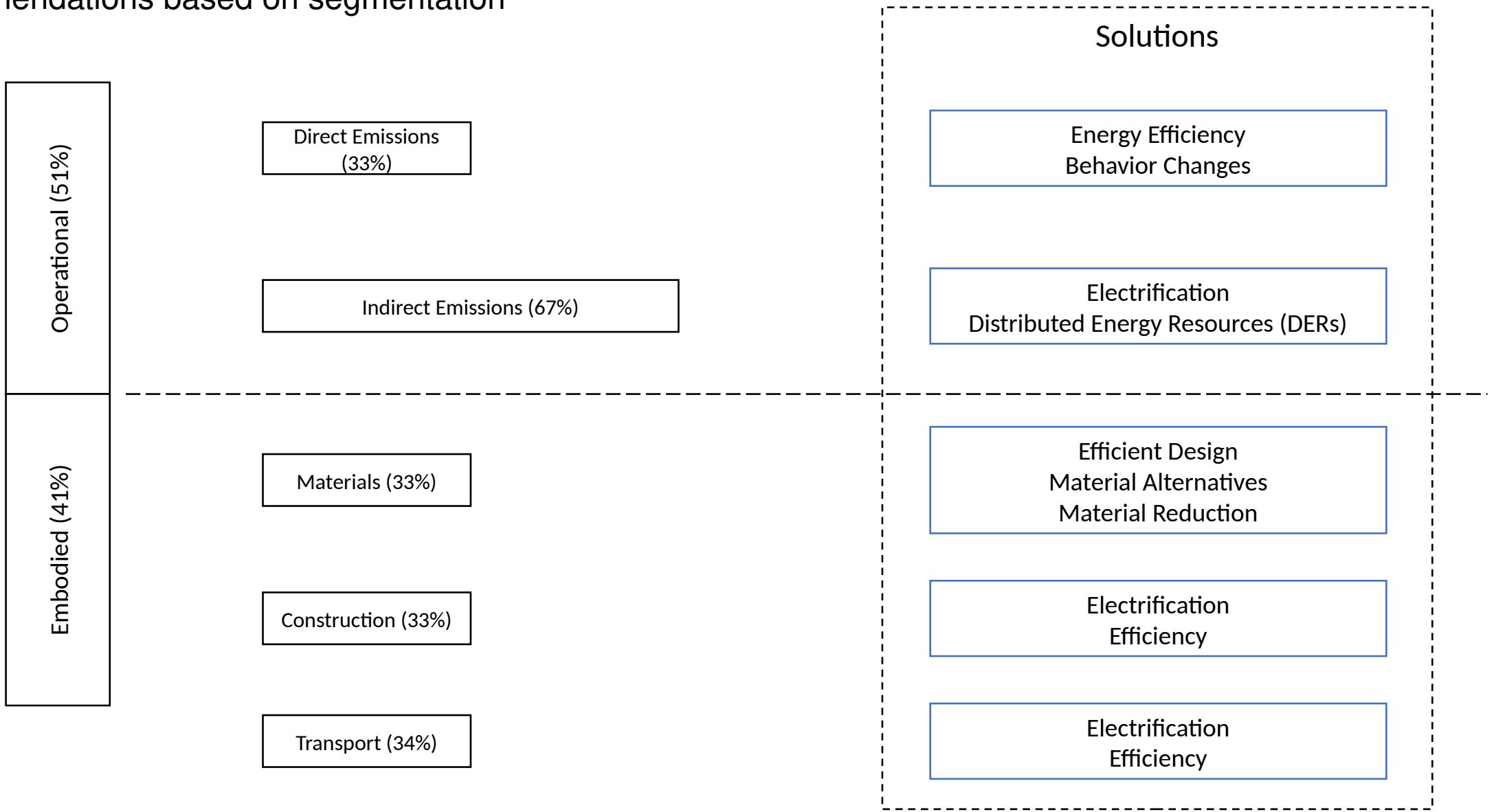


Source: ASHRAE

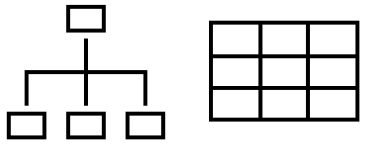
Decarbonization intervention at each stage of the building lifecycle



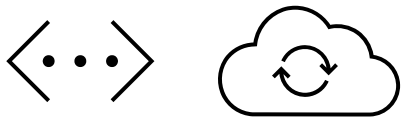
Recommendations based on segmentation



Backed by data science and



Structured
Time Series Data
GPS Data
System Logs
CRM/ERP Data
Booking Systems

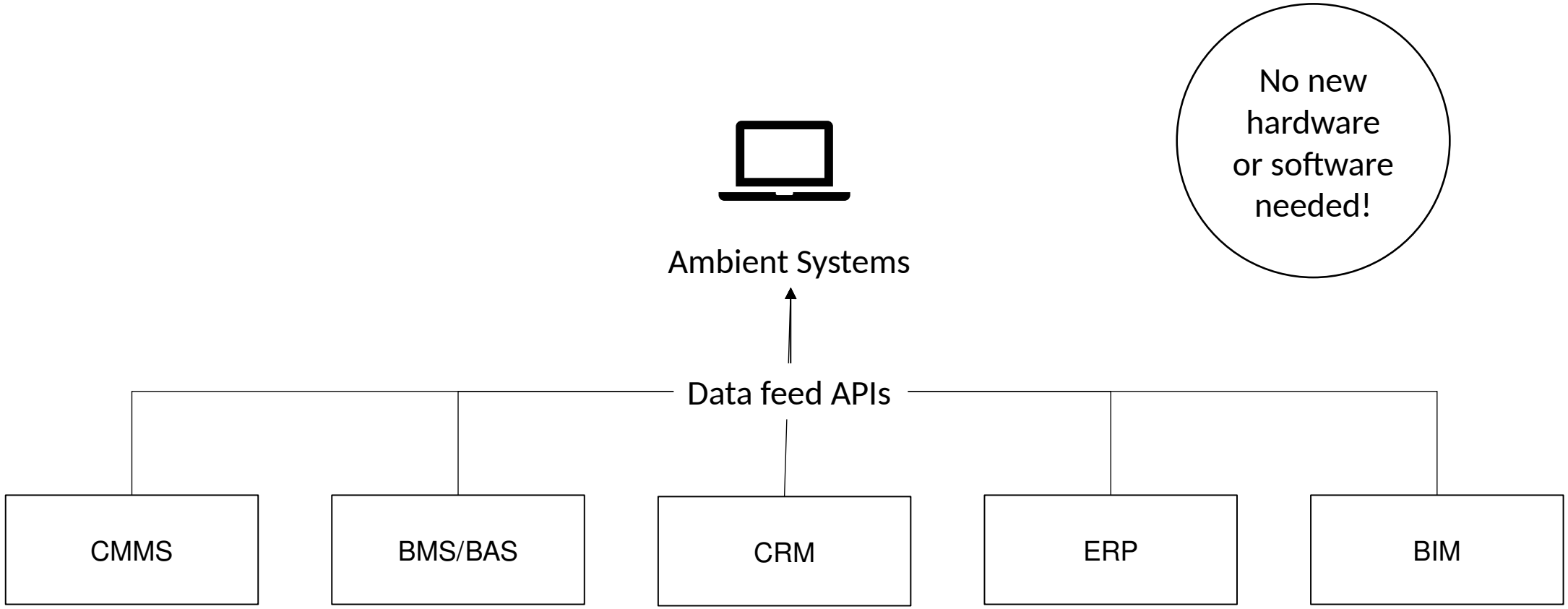


Semi-Structured Data
Emails
HTML/XML files
JSON files
Network Traffic

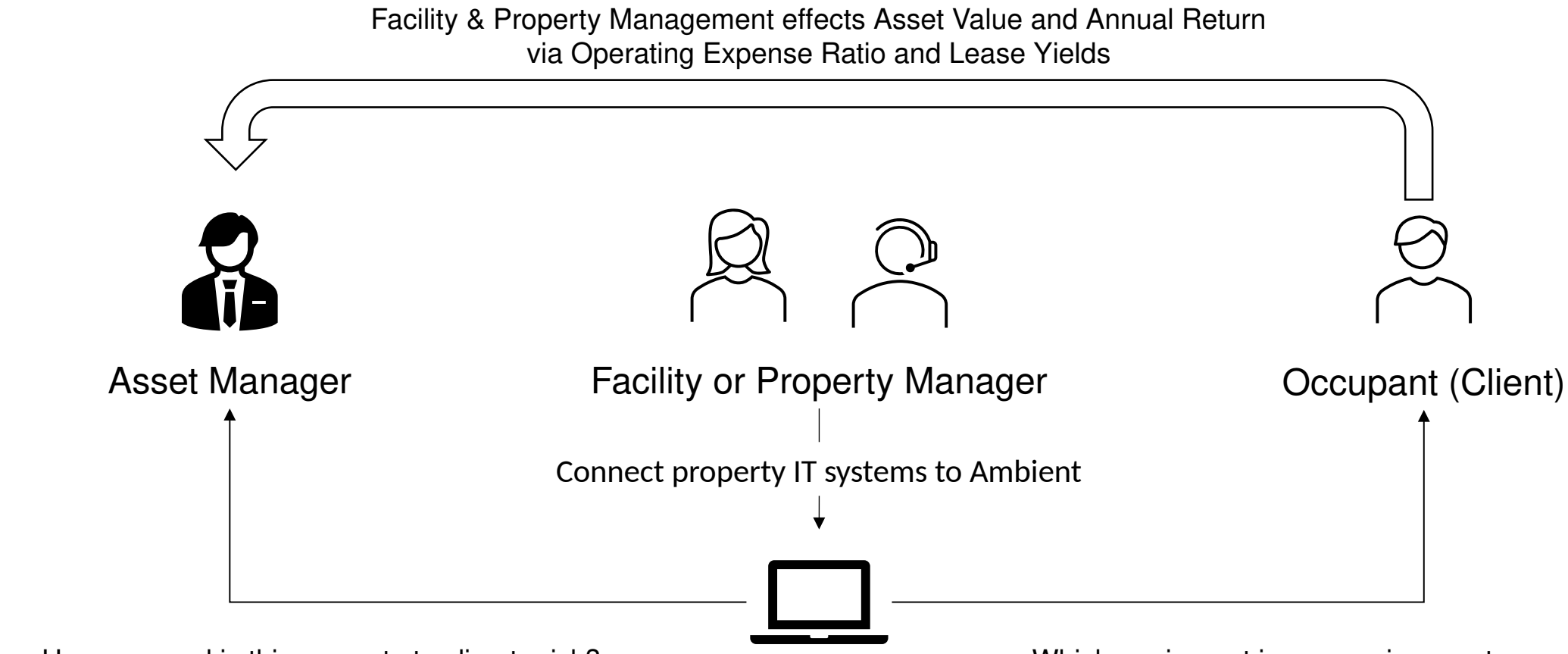


Unstructured
Text
Audio
Video
Images

Build on your existing platform infrastructure



Insights: Ambient provides recommendations on how to improve property valuation & profitability



Key Features

Ambient Offers 7 Main Features

Feature	Category	Sub Feature
1	User Management	a) User Registration b) User Sign-in
2	Objective Selection	**a) Select Key Metrics**
3	Data Input	a) Select Data Feeds **b) Connect Data Feeds** c) Data E d) Data E hment
4	Data Visualization	
5		**a) Query Processing** **b) Relation Discovery** c) Knowledge Graph **d) Multi-objective Optimization**
6	Output Delivery	a) Data Visualization in a Web App b) Text Message **c) Email (Text format or PDF Report)**
7	Controls	**a) BMS/BAS Automation**

Ambient provides recommendations based on customer's unique goals

Commercial Real Estate Lifecycle

Asset Managers & Executives

- Net operating income (NOI)
- The Capitalization Rate
- Sinking fund factor (SFF)
- Gross Income Multiplier
- Debt-service coverage ratio (DSCR)
- IRR

Property Managers & Brokers

- Lease information: critical lease events, active vs. inactive leases, and lease details
- Occupancy and Vacancy Rates
- Space management: space utilization, occupational density
- Tenant Feedback: satisfaction with environment, indoor comfort, turnover

Engineering & Facilities Managers

- Repairs: work orders, time to resolve, quality (time until repeat fix)
  ical, gas, fire
  : scheduled maintenance,
  maintenance, costs
- Utilities: settings

FM effects Asset Value and **Annual Return** via **Operating Expense Ratio**

Ambient can process structured and unstructured data through its data engineering pipeline

Stakeholders	Dataset	Platform	Vendors	Data Format
Asset Managers & Executives	Financial Model	Excel	Microsoft	Tables
	Financial data	Accounting software	Intuit QuickBooks, Xero	Tables
	ESG	Carbon accounting software	Persefoni	Tables
	Workforce HR Data	ERP	SAP, Oracle, Microsoft Dynamics	Tables
Property Managers & Brokers	Tenant Lease information	CRM	Salesforce, SAP, VTS	Tables, Text
	Market Data	Internal software managed by individual brokers	RE/MAX Commercial, Savills, CBRE, Colliers, Cushman & Wakefield	Tables, Vision
Engineers & Facility Managers	Structural, Mechanical, Health & Safety	BIM	AutoCAD	Tables, Vision
	Life Cycle Assessment	LCA tool	EC3, Athena, BEES, Tally	Tables, Vision
	AC and Electrical	BMS / BAS	Siemens Building Techn Honeywell, Bosch, ABB,	,
		Computerized maintenance management system (CMMS)	Facilities Management eXpress (FMX), UpKeep, Fiix, Fracttal, Corrigo	Tables, Vision

Ambient let’s you get the simple answers to key questions

How do we improve indoor air quality while saving energy costs?



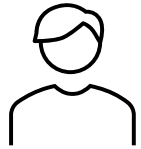
Facility Manager

Should we invest in new technology to reduce emissions?



Asset Manager

Which equipment is consuming the most energy?

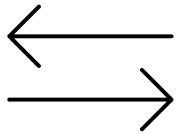


Occupant / Tenant

Ambient identifies mapping between key independent and dependent variables

Independent Variables

Local Supporting Infrastructure	**New**	**Ambient**
Building Technologies	**New**	
Climate Change Physical Risk	**Climate**	
Climate Change Transition Risk	**Climate**	
Employment Growth in Related Sectors	Conventional	**Baseline**
Employment Rate	Conventional	
Economic GDP Growth	Conventional	
New construction supply	Conventional	
Inflation	Conventional	
Facility Management Costs	Conventional	
Building Construction Type	Conventional	
Property Location	Conventional	



Dependent Variables

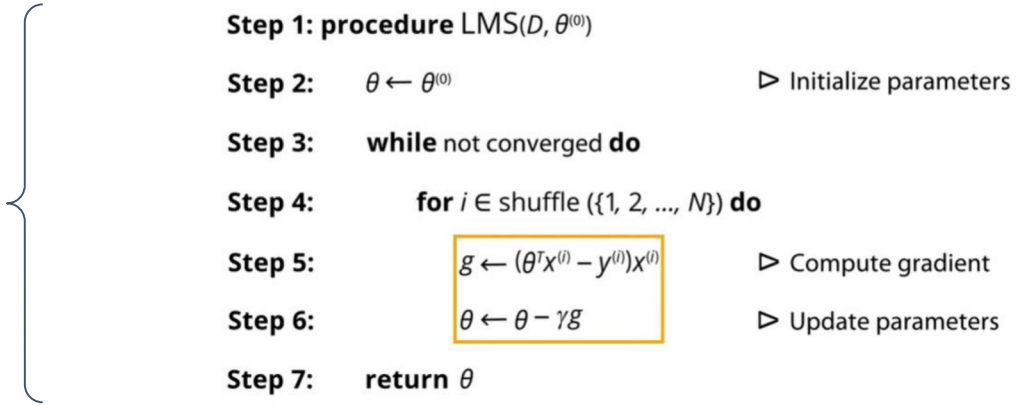
1. Asset Valuation & Appraisal Confidence
2. Occupancy / Vacancy
3. Operating Expense and Operating Expense Ratio

Ambient calculates trade-offs among competing priorities to provide balanced recommendations

Key Steps:

1. Identify variable correlations (e.g., Energy Consumption and Temperature)
2. Formulate objective function (e.g., minimize energy consumption)
3. Formulate constraints functions (e.g., IAQ, CO2, Relative Humidity, Thermal Comfort)
4. Optional: Assign objective vector weights
5. Perform multi-objective optimization leveraging (i.e., vector optimization)

Stochastic Gradient Descent



Ambient sends you key daily summary so you can stay on track

Daily Email

To: Facility Management Team
cc: Asset Manager, Property Manager
From: Ambient Team
Subject: Today's Plan

We have scheduled adjustments to the temperature and ventilation based on lower occupancy.

You will need to make sure windows are not open.

We estimate that daily energy and water savings will be about $500 based on these recommendations. This is better than your average daily consumption in the past month.

We also identified that AC duct is due for maintenance. Check in on it to make to avoid equipment downtime.

Weekly Email

Monthly Email

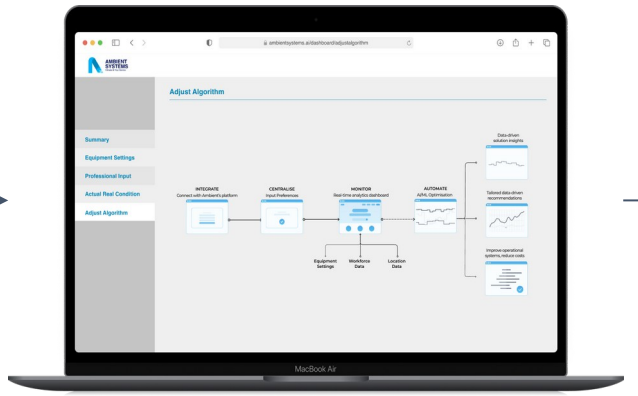
To: Asset Manager
cc: Property Manager
From: Ambient Team
Subject: Next Month's Strategy

Your portfolio is performing better than before.

We recommend looking at these top three emitting properties:
1)
2)
3)

Ambient connects with your BMS or BAS to implement recommendations

Ambient helps your facility managers make se
data and streamline



BMS Database

Siemens Building Technologies, , Honeywell, Bosch, ABB,
Sch

Extra Slides

Appendix

Extending Traditional Real Estate Metrics to support Decarbonization

Macroeconomic Outlook	Property Performance	Property Features
Medium	Vacancy Rate	Year Built

Traditional Variables from Existing Operations Provide Baseline

+Facility Use	+Local Infrastructure	+Facility Management Features
Tenant Industry Outlook Industry Climate Resilience	Points of interest (e.g., 4 start hotels nearby) Transportation Accessibility Energy Stability	+ FM Needs & Capabilities + Digital Technology + Equipment

New Operational Variables from External and Internal Sources

+Transition Risk	+Physical Risk
Carbon-Intensive asset Local Energy and Carbon Standards Lower emission building materials	Physical Hazard (e.g., floods and earthquakes) Increased maintenance cost Investment required to improve building facilities



Climate Risk Data from Global Insurance Leader

Examining a range of property output metrics

Baseline RE Investment Metrics

- Net operating income (NOI)
- The Capitalization Rate
- Sinking fund factor (SFF)
- Gross Income Multiplier
- Debt-service coverage ratio (DSCR)
- IRR

Variables

Predicted vs Actual Income
Forecast demand, lease price, and occupancy

Predicted vs Actual
Predicted vs Actual on

Investment Management

Enhancing Valuation Accuracy
+
Manage Climate Risk

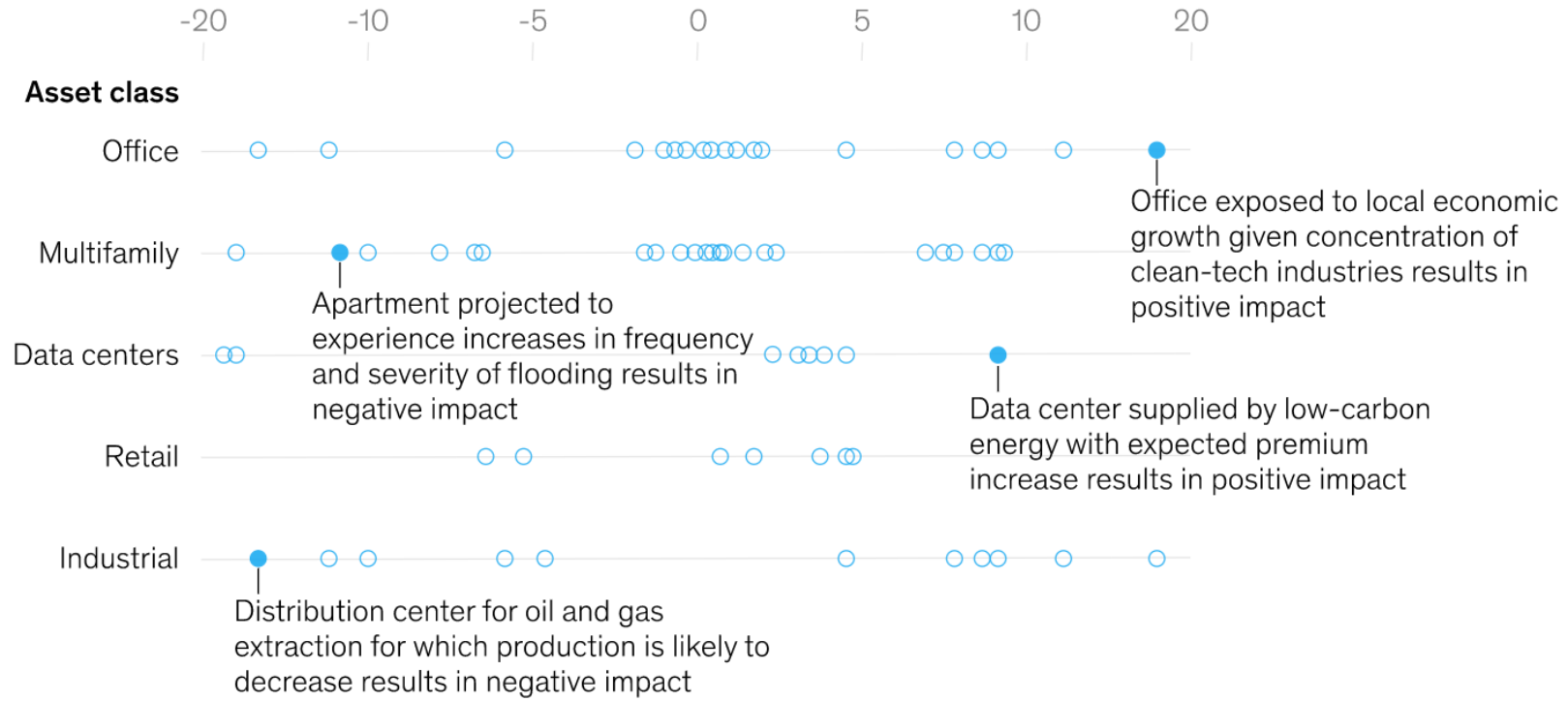
Asset Management

Lowering

Balance with Operating Expense Ratio
+
Income & Valuation

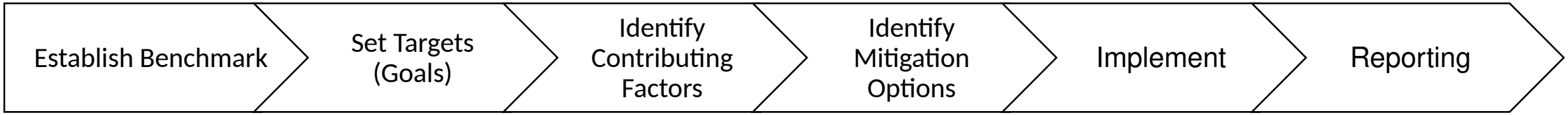
Property level data mapped to portfolio wide impacts

Illustrative chart and examples of physical and transition risk effects on equity value of assets, %



McKinsey & Company

Decarbonization Process



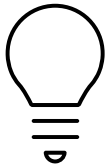
Carbon Mitigation Levers

Portfolio Level
- Diversification
- Divestment
- Mitigation and Offsetting

Property Level Optimization
- Managing Demand and Consumption
- Low Carbon Development & Construction
- Retrofits for Efficiency
- Upgrades to AC or Heating
- Upgrades to Lighting

Enhancing operational efficiency and competitive position

Enhance Operational Efficiency	Enhance Competitive Position
• Reduce FM, technician, and skilled trades costs • Improve Energy Efficiency • Enhanc • Reduce (Operational & Embodied) • Improve • Enhance asset yields and profitability	• Portfolio investment performance • Quality of service provided to occupiers • Investor liquidity and willingness to invest • Fund terms (including fees) • Brand recognition • Business reputation



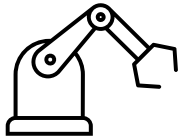
15% to 25%

Energy Savings



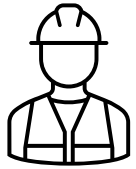
20% to 40%

Reduction in



10% to 20%

Increase in Equipment Uptime



3X

Manhour Time Savings

Part 1 – Platform: tailored CRE data and insights platform, balancing productivity with sustainability

Insights



Portfolio

- Portfolio Performance
- Leasing
- Transactions



Facilities & Operations

- Decarbonization
- erformance & Repairs
- ncy



Space & Workplace

- Workplace Performance
- Occupancy
- Utilization

Platform

CRE Data Model	API Connections	Data Engineering	Data Visualization	Notifications